Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Databricks Lakehouse Architecture, Cloud Data Modernization, Enterprise Data Foundations, Delta Lake & Workflows, Dimensional Data Modeling, Star/Snowflake Schemas, SCD Type 1/2, Batch and Streaming Data Workflows, Data Quality & Governance Frameworks, Predictive Modeling & Analytics Integration

Programming Languages & Tools: Databricks, Apache Spark, PySpark, Python, Structured Query Language (SQL), Azure Data Factory, Azure Synapse, Azure SQL, AWS (S3, Redshift), Snowflake, Power BI, Tableau, Airflow

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Databricks Data Engineer — **May 2022 – Present**

- **Architected high-throughput data processing pipelines** utilizing Databricks, Apache Spark, PySpark, and advanced SQL stored procedures to modernize enterprise data platforms and enable large-scale transformation.
- **Implemented robust dimensional data modeling patterns** including Star/Snowflake schemas and Slowly Changing Dimensions (SCD Type 1/2) within Azure Cloud and Databricks lakehouse environments to improve data quality and scalability.
- **Designed automated data quality audit frameworks** with firewalls and proactive anomaly detection machine learning systems, achieving 95% accuracy and reducing operational turnaround by over 80%.
- **Engineered advanced machine learning and predictive workflows** using XGBoost, Random Forests, and SARIMA time-series models integrated with Databricks to support customer behavior analytics and optimize workforce capacity planning by 20%.
- **Maintained 99.9% data availability SLAs** by establishing production runbooks, Airflow batch monitoring, and comprehensive Root Cause Analysis (RCA) postmortems across complex cloud data ecosystems.

Graduate Data Science Intern — **May 2022 – Present**

- **Developed scalable predictive modeling algorithms** leveraging Python, Scikit-Learn, and SQL to analyze complex datasets and improve operational reliability.
- **Built interactive executive dashboards** using Power BI (DAX, measures) and Tableau to translate large-scale data processing outcomes into actionable business intelligence.
- **Collaborated with cross-functional technical teams** to prototype data ingestion and transformation pipelines, adhering to enterprise data governance standards.

Engineering Graduate Trainee — **May 2022 – Present**

- **Optimized database query performance** and engineered backend data processing routines to support high-volume industrial and operational analytics.
- **Automated data validation procedures** using scripting and SQL to streamline engineering workflows and eliminate manual reporting errors.
- **Partnered with senior engineers** to troubleshoot system integration issues and document technical specifications for data pipelines.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.

- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.